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RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

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Department of Computer Engineering

Semester: III **Academic Year 2026-27 (ODD)**

Assignment 1

Course Code: 2413CEC3T2
Course Name: DATABASE MANAGEMENT SYSTEMS
Max. Marks: 10 **Submission Date: / /2026**

Question	Solve all the questions	Max. Marks	CO	BT
Q1. A)	Explain Specialization and Generalization with suitable example	3	CO1	BT2
Q 1. B)	Draw ER model for a Library Management System A library maintains books, members, and librarians. Each book has Book_ID, Title, Author, and Price. Each member has Member_ID, Name, and Contact. A member can borrow many books. A book can be borrowed by different members over time. Each librarian issues books to members.	2	CO1	BT4
Q 2. A)	Explain fundamental operations of relational algebra with suitable examples- i) Select (σ) ii) Project (Π) iii) Union (∪) iv) Set Difference (-) v) Cartesian product (X)	3	CO2	BT2
Q 2. B)	Consider the following relational database and construct the relational algebra queries for the following: Account (Branch_name, Account_number, Balance) Loan (Branch_name, Loan_number, Amount) Depositor (Cust_name, Account_number) Borrower (Cust_name, Loan_number) 1) Find the names of customers who have account at the bank as well as borrowed a loan from the bank. 2) Find the customer names who have account at Roadridge branch. 3) Find the account balances at Perryridge branch. 4) Find the customer names whose loan amount is greater than 50000. 5) Find the customers who have account but not loan at the bank.	2	CO2	BT3

Course Outcomes (CO) -Learner will be able to:

CO1: Understand the fundamentals of database systems and design ER for the real-life problem.
CO2: Convert conceptual model to relational model and formulate relational algebra queries.
CO3: Design and querying database using SQL.
CO4: Analyze and apply concept of normalization to relational database design.
CO5: Implement triggers, stored procedures and functions in PL/SQL.
CO6: Understand query processing and query optimization. Understand the concept of transactions, concurrency, recovery management and query optimization.

Bloom's Taxonomy:

BT1- Remembering, BT2- Understanding, BT3- Applying, BT4- Analyzing, BT5- Evaluating, BT6- Creating